API Monitoring Documentation: Error Rate & Status Code Monitoring

1. Introduction

Monitoring is a critical part of any production system. For the Bookstore API, it is essential to track how the application behaves in real time and identify failures quickly.

This document explains how monitoring is implemented using:

- **Status Code Monitoring**
- **Error Rate Monitoring**
- **Dashboards and Alerts**

The goal is to improve visibility, detect failures early, and ensure system reliability.

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2. Objectives

- Monitor API health and performance
- Track success and failure responses
- Detect abnormal behavior in real time
- Enable faster troubleshooting using dashboards and alerts

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3. Key Monitoring Concepts

3.1 Status Code Monitoring

HTTP status codes indicate how the server responds to each request.

🔹 **Categories:**

Status Code	Meaning	Example
2xx	Success	200 OK
4xx	Client Error	404 Not Found
5xx	Server Error	500 Internal Server Error

Purpose of Status Code Monitoring

- Understand API behavior in detail

- Identify whether errors are from client or server
- Track trends of different response types

Visualization Used

- Line chart → Response trends over time
- Pie chart → Distribution of status codes
- Stacked graph → Comparison of 2xx, 4xx, 5xx

3.2 Error Rate Monitoring

Error rate represents the percentage of failed requests.

Formula

Error Rate = (Failed Requests / Total Requests) × 100

What is considered an error?

- 4xx responses (client errors)
- 5xx responses (server errors)

Purpose of Error Rate Monitoring

- Provide a quick health overview
- Detect sudden spikes in failures
- Trigger alerts for abnormal conditions

Visualization Used

- Line chart → Error trend over time
- Single stat → Current error percentage

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4. Difference Between Error Rate and Status Code Monitoring

Feature	Status Code Monitoring	Error Rate Monitoring
Type	Detailed	High-level
Purpose	Diagnose issues	Detect overall health

Output	2xx, 4xx, 5xx breakdown	% of failed requests
Use Case	Root cause analysis	Alerting & monitoring

Simple Explanation

- Status Codes → *What exactly is happening?*
- Error Rate → *How bad is the situation?*

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5. Dashboard Design

The monitoring dashboard is designed to be:

- Clean and easy to understand
- Organized for quick troubleshooting
- Focused on key metrics

Dashboard Components

1. **Total Request Count**
2. **2xx Responses (Success)**
3. **4xx Responses (Client Errors)**
4. **5xx Responses (Server Errors)**
5. **Error Rate (%)**
6. **Response Trend Over Time**

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6. Alerts Configuration

Alerts are configured to notify when abnormal behavior occurs.

Alert Conditions

Alert Type	Condition
High 5xx Error	> 5% server errors
High 4xx Error	Sudden increase in client errors
High Error Rate	> 10% failure rate

Low Success Rate	Drop in 2xx responses
API Down	No response / unhealthy

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7. Monitoring Workflow

1. API receives requests
2. Metrics are collected (status codes, counts)
3. Dashboard visualizes data
4. Alerts trigger on abnormal patterns
5. Team investigates using dashboard insights

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8. Benefits of This Monitoring Setup

- Faster issue detection
- Clear visibility into API behavior
- Reduced downtime
- Better debugging using detailed insights
- Improved reliability and performance

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9. Conclusion

Both monitoring techniques play an important role:

- **Status Code Monitoring** provides detailed insights into API responses
- **Error Rate Monitoring** gives a quick overview of system health

Together, they ensure efficient monitoring, faster troubleshooting, and better system reliability.

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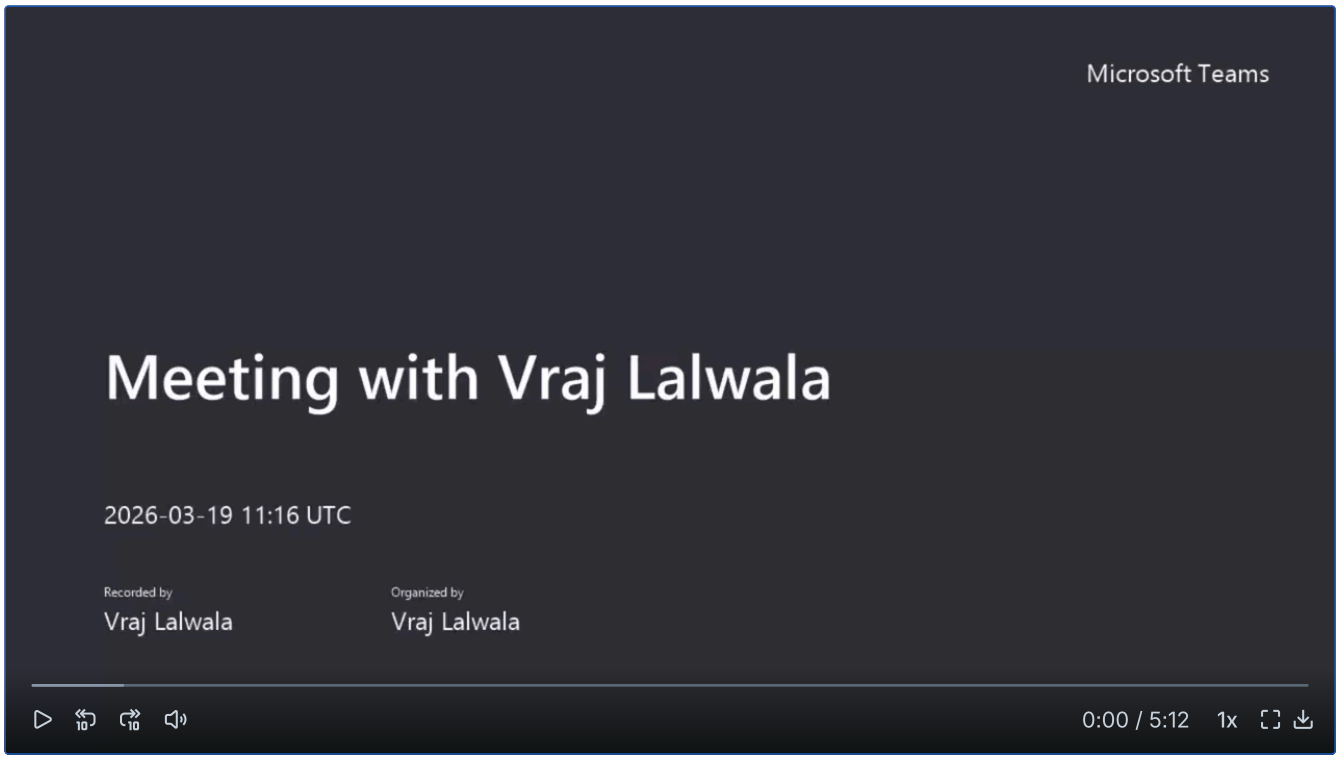
10. Video Explanation



This video explains:

- Error Rate Monitoring
- Status Code Monitoring
- Key differences and use cases